

The unit circle



1 State the quadrant in which the following angles are located:

- | | | | |
|----------------------|----------------------|-----------------------|-----------------------|
| a 299° | b 5° | c 160° | d 229° |
| e 40° | f 310° | g -138° | h -244° |

2 For the given functions, state the following:

- i The quadrants where the function is positive.
- ii The quadrants where the function is negative.

- | | |
|---------------------------|--------------------------|
| a Sine function | b Cosine function |
| c Tangent function | |

3 State whether the values of the following are positive or negative:

- | | | | |
|---------------------------|--------------------------|---------------------------|---------------------------|
| a $\sin 310^\circ$ | b $\sin 50^\circ$ | c $\sin 130^\circ$ | d $\sin 230^\circ$ |
| e $\sin 31^\circ$ | f $\tan 31^\circ$ | g $\cos 267^\circ$ | h $\sin 267^\circ$ |
| i $\cos 180^\circ$ | | | |

4 Consider the given angles below and state whether the following are true or false:

$$w = 253^\circ, \quad x = 265^\circ, \quad y = 193^\circ, \quad z = -258^\circ, \quad a = -182^\circ, \quad b = -257^\circ$$

- a** Angles z , a , and b are in Quadrant 3.
- b** $\sin z$ and $\sin b$ are both negative.
- c** $\tan(z - 90^\circ)$ and $\tan(z + 90^\circ)$ are both positive.
- d** The angles w , x , and y have negative cosine values.
- e** Angles w , x , and y are in Quadrant 3.

5 State the quadrant where the point in each scenario is located:

- a** The point (x, y) is on the unit circle at a rotation of θ such that $\sin \theta = \frac{5}{13}$ and $\cos \theta = \frac{12}{13}$.
- b** The point $P(x, y)$ is on the unit circle at a rotation of θ such that $\sin \theta = -\frac{12}{37}$ and $\cos \theta = \frac{35}{37}$.

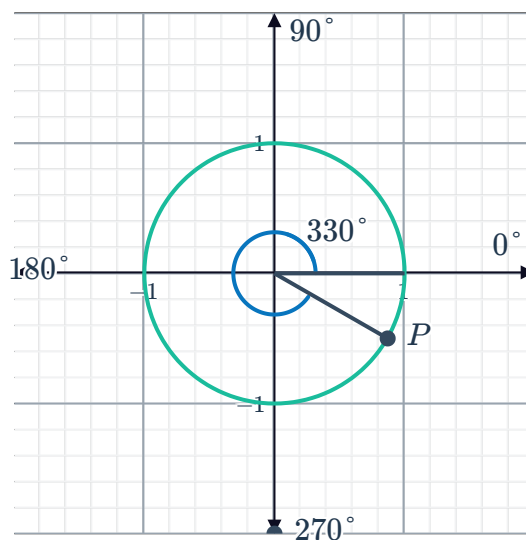
6 State the quadrant where the angle in each scenario is located:



- a θ is an angle such that $\sin \theta > 0$ and $\cos \theta < 0$.
- b θ is an angle such that $\tan \theta < 0$ and $\sin \theta > 0$.
- c θ is an angle such that $\tan \theta < 0$ and $\cos \theta < 0$.
- d θ is an angle such that $\tan \theta > 0$ and $\sin \theta > 0$.

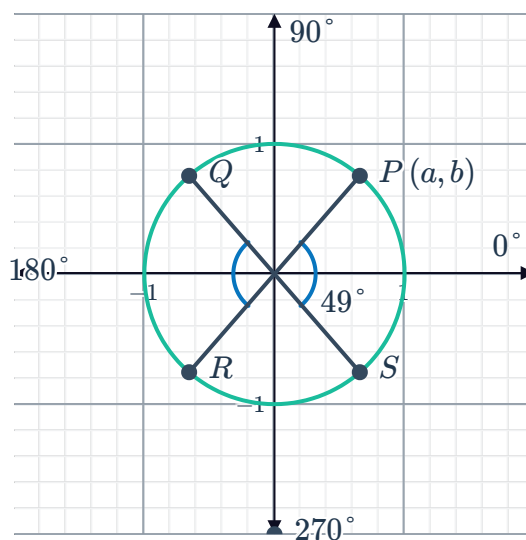
Equivalent angles

7 Point P on the unit circle shows a rotation of 330° . Find the related acute angle in the first quadrant.



8 The diagram shows points $P(a, b)$, Q , R and S , which represent rotations of 49° , 131° , 229° and 311° respectively around the unit circle.

- a Find the coordinates of the following points in terms of a and b :
 - i Q ii R iii S
- b Find the equivalent trigonometric ratio in the first quadrant of the following:
 - i $\sin 131^\circ$ ii $\sin 229^\circ$
 - iii $\sin 311^\circ$
- c Hence, express each of the following in terms of $\sin x$:
 - i $\sin (180^\circ - x)$ ii $\sin (180^\circ + x)$
 - iii $\sin (360^\circ - x)$



- 9 The diagram shows points $P(a, b)$, Q , R and S which represent rotations of 63° , 117° , 243° and 297° respectively around the unit circle.

- a Find the coordinates of the following points in terms of a and b :

i Q ii R iii S

- b Find the equivalent trigonometric ratio in the first quadrant of the following:

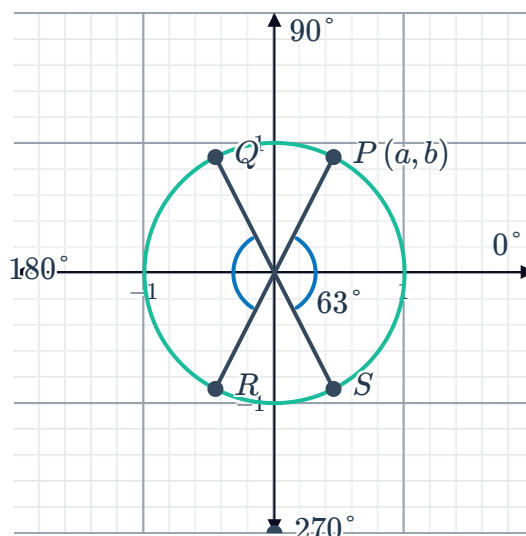
i $\cos 117^\circ$ ii $\cos 243^\circ$

iii $\cos 297^\circ$

- c Hence, express each of the following in terms of $\cos x$:

i $\cos(180^\circ - x)$ ii $\cos(180^\circ + x)$

iii $\cos(360^\circ - x)$



- 10 The diagram shows points $P(a, b)$, Q , R and S , which represent rotations of 62° , 118° , 242° and 298° respectively around the unit circle.

- a Find the coordinates of the following points in terms of a and b :

i Q ii R iii S

- b Find the equivalent trigonometric ratio in the first quadrant of the following:

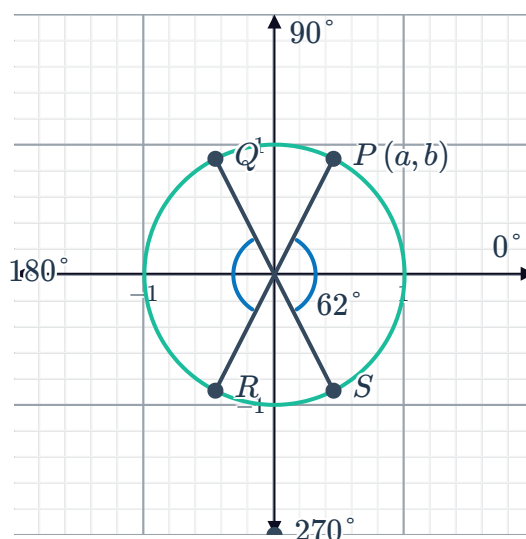
i $\tan 118^\circ$ ii $\tan 242^\circ$

iii $\tan 298^\circ$

- c Hence, express each of the following in terms of $\tan x$:

i $\tan(180^\circ - x)$ ii $\tan(180^\circ + x)$

iii $\tan(360^\circ - x)$



11 For each of the following graphs, find:

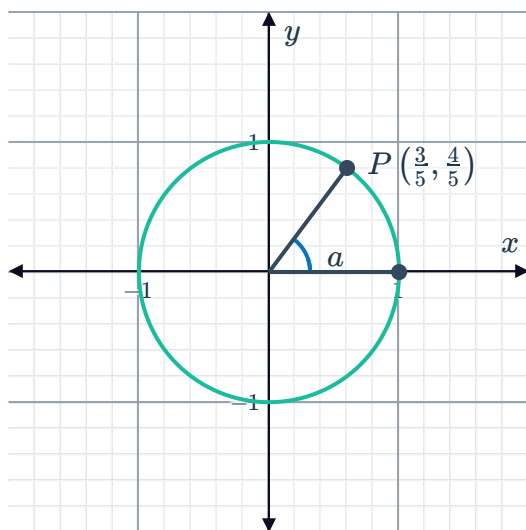


i $\sin a$

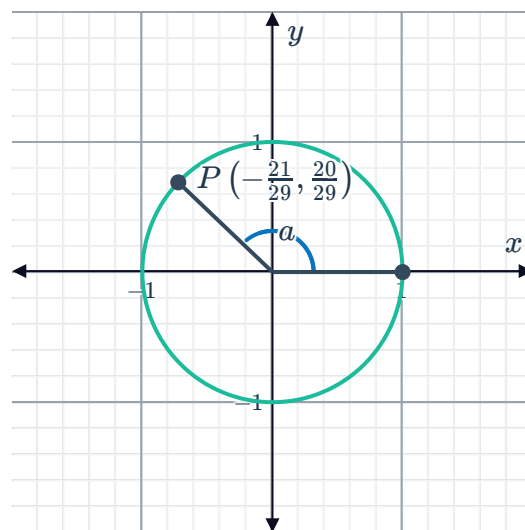
ii $\cos a$

iii $\tan a$

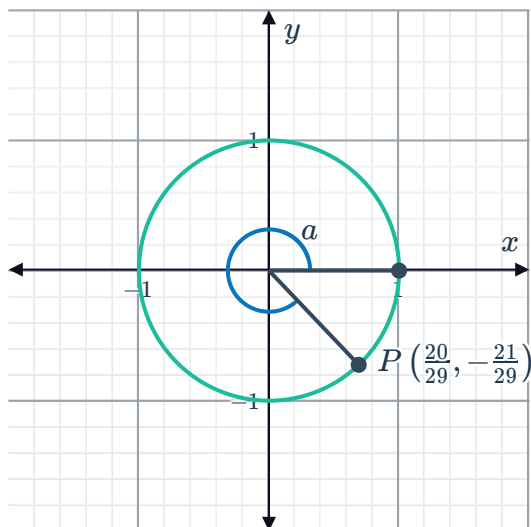
a



b



c



12 The following are angle rotations at point P on the unit circle. Find the acute angle in the first quadrant related to each rotation:

a 295°

b -150°

c 240°

d -75°

13 Rewrite each expression as a trigonometric ratio of a positive acute angle:

a $\sin 93^\circ$

b $\cos 195^\circ$

c $\tan 299^\circ$

d $\sin(-139^\circ)$

e $\cos(-222^\circ)$

f $\tan(-289^\circ)$

14 Simplify:



- a** $\sin(360^\circ + \theta)$ **b** $\cos(360^\circ - \theta)$ **c** $\sin(180^\circ - \theta)$ **d** $\cos(180^\circ + \theta)$
e $\sin(180^\circ + \theta)$ **f** $\sin(90^\circ - \theta)$ **g** $\sin(-\theta)$ **h** $\cos(-\theta)$

15 Given the approximations $\cos 21^\circ = 0.93$ and $\sin 21^\circ = 0.36$, find the approximate values of the following and giving your answers in two decimal places:

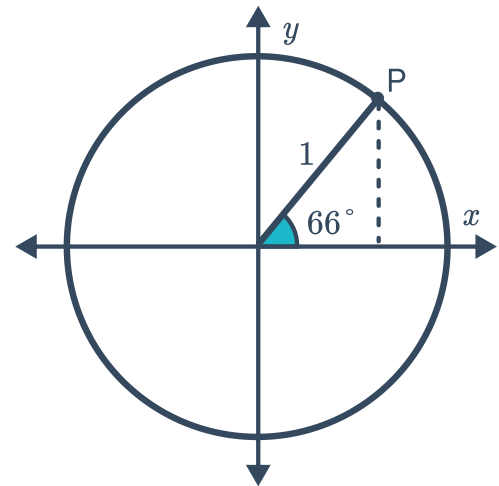
- a** $\cos 339^\circ$ **b** $\sin(-339^\circ)$ **c** $\cos 159^\circ$ **d** $\sin 159^\circ$
e $\sin 201^\circ$ **f** $\cos 201^\circ$ **g** $\sin(-201^\circ)$ **h** $\cos(-201^\circ)$

16 Evaluate the following correct to two decimal places:

- a** $\sin 146^\circ$ **b** $\tan 386^\circ$ **c** $\cos 387^\circ$

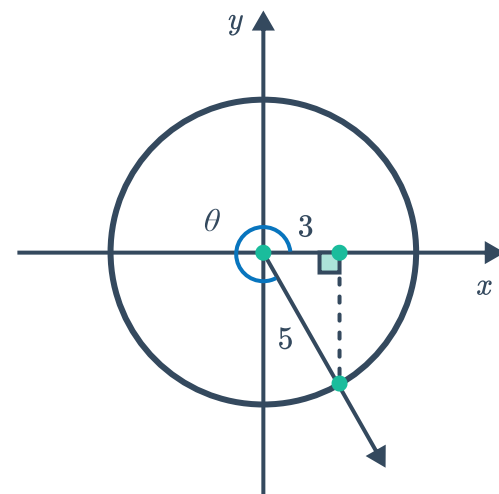
17 The diagram shows P , which represents a rotation of 66° around the unit circle:
Find the following to two decimal places:

- a** Coordinates of $P(x, y)$
b The new coordinates of P if it was reflected about the y -axis.
c $\sin 114^\circ$
d $\cos 114^\circ$
e $\tan 114^\circ$



18 Suppose that $\cos \theta = \frac{3}{5}$, where $270^\circ < \theta < 360^\circ$.

- a** Find $\sin \theta$. **b** Find $\tan \theta$.

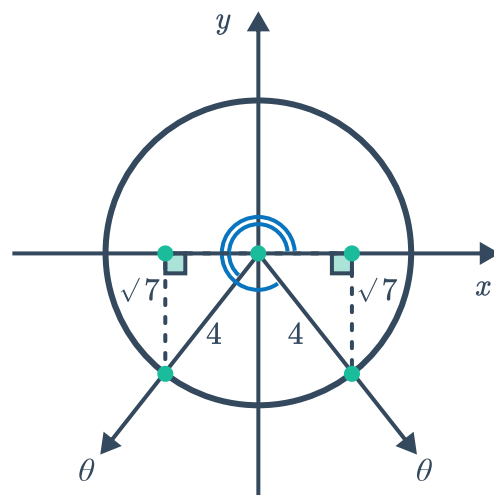


19 Suppose that $\sin \theta = -\frac{\sqrt{7}}{4}$.



a Find $\cos \theta$.

b Find $\tan \theta$.



20 Consider an angle θ such that $\sin \theta = 0.6$ and $\tan \theta < 0$, find:

a $\sin \theta$

b $\cos \theta$

c $\tan \theta$

21 Consider an angle θ such that $\tan \theta = -1.3$ and $270^\circ < \theta < 360^\circ$, find:

a $\cos \theta$

b $\sin \theta$

22 Given that $\tan \theta = -\frac{15}{8}$ and $\sin \theta > 0$, find $\cos \theta$.

23 Given the following, find $\sin \theta$.

a $\cos \theta = -\frac{60}{61}$ and $0^\circ \leq \theta \leq 180^\circ$

b $\cos \theta = -\frac{6}{7}$ and $\tan \theta < 0$

24 Given the following, find the value(s) of $\tan \theta$.

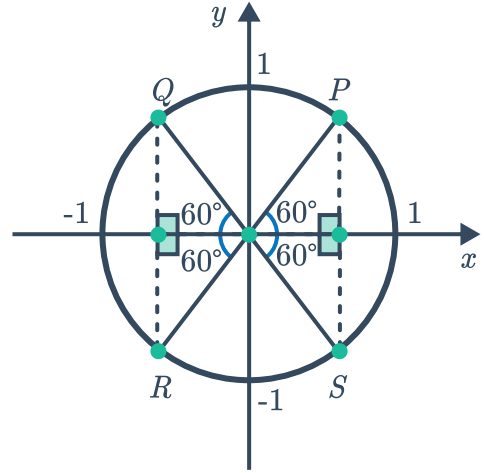
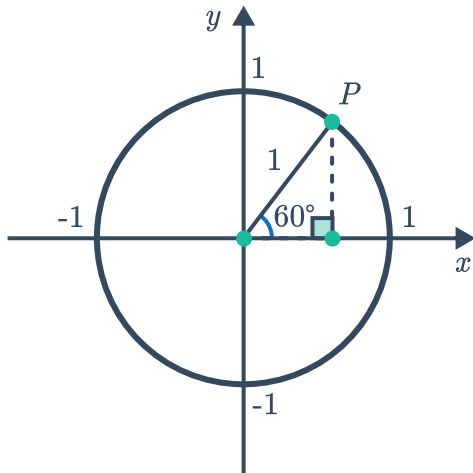
a $\cos \theta = \frac{3}{7}$ and θ is acute

b $\sin \theta = \frac{1}{\sqrt{10}}$ and $-90^\circ \leq \theta \leq 90^\circ$



Exact trigonometric ratios

- 25** The first diagram shows a unit circle with point $P \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right)$ marked on the circle. Point P represents a rotation of 60° anticlockwise around the origin from the positive x -axis:

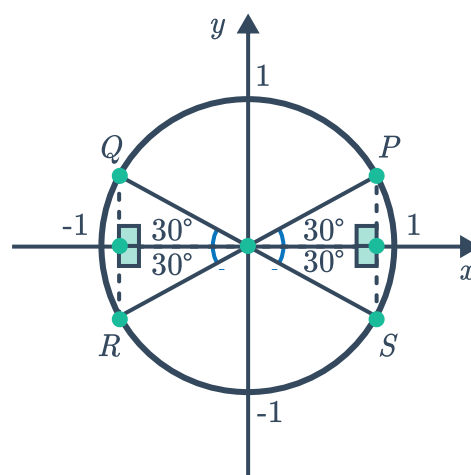
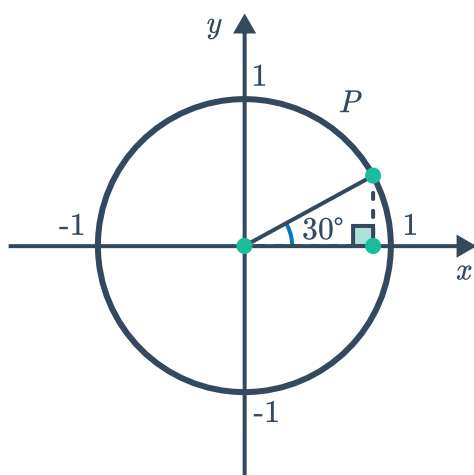


- a** Find the exact values of the following:
- i** $\sin 60^\circ$ **ii** $\cos 60^\circ$ **iii** $\tan 60^\circ$
- b** On the second diagram, the coordinate axes shows a 60° angle that has also been marked in the second, third, and fourth quadrants. For each quadrant, find the relative angle.
- i** Quadrant 2 **ii** Quadrant 3 **iii** Quadrant 4
- c** The points Q , R and S mark rotations of point P to the corresponding angles on the unit circle. Find the exact coordinates of each point:
- i** Q **ii** R **iii** S

26



The first diagram shows a unit circle with point $P \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right)$ marked on the circle. Point P represents a rotation of 30° anticlockwise around the origin from the positive x -axis:



a Find the exact values of the following:

- i $\sin 30^\circ$ ii $\cos 30^\circ$ iii $\tan 30^\circ$

b On the second diagram, the coordinate axes shows a 30° angle that has also been marked in the second, third and fourth quadrants. For each quadrant, find the relative angle:

- i Quadrant 2 ii Quadrant 3 iii Quadrant 4

c The points Q , R and S mark rotations of point P to the corresponding angles on the unit circle. Find the exact coordinates of each point:

- i Q ii R iii S

27 Consider the trigonometric ratio $\cos(-210^\circ)$.

- a Find the value of the related acute angle.
b Hence, find the value of $\cos(-210^\circ)$ using exact values.

28 Find the exact value of the following:

- | | |
|--|------------------------------------|
| a $\sin 225^\circ$ | b $\cos 225^\circ$ |
| c $\tan 225^\circ$ | d $\sin 690^\circ$ |
| e $\sin 135^\circ$ | f $\cos 315^\circ$ |
| g $\tan 150^\circ$ | h $\sin 240^\circ$ |
| i $\cos 690^\circ$ | j $\tan 690^\circ$ |
| k $\sin 50^\circ + \sin 160^\circ + \sin 200^\circ + \sin 310^\circ$ | l $\tan 60^\circ + \tan 315^\circ$ |



29 By rewriting each ratio in terms of the related acute angle, find the exact value of:

$$\frac{\sin 120^\circ \cos 240^\circ \tan 330^\circ}{\tan (-45)^\circ}$$